

1. Compilers

A compiler translates the **entire** source code into an executable file (like an `.exe`) before the program ever runs.

- **Process:** Scanning → Analysis → Optimization → Code Generation.
- **Advantages:** Faster execution (the translation is already done) and better optimization.
- **Disadvantages:** Slower development cycle (you must re-compile after every change).
- **Examples:** C, C++, Rust, Go.

2. Interpreters

An interpreter translates and executes the code **line-by-line** or statement-by-statement.

- **Process:** It reads a line, converts it to intermediate code/machine code, and executes it immediately.
- **Advantages:** Easier debugging (stops exactly where the error is) and highly portable.
- **Disadvantages:** Generally slower execution because the translation happens while the program is running.
- **Examples:** Python, Ruby, PHP

Summary Comparison Table

Feature	Compiler	Interpreter
Execution	Translates all at once	Translates line-by-line
Output	Generates an intermediate object code	No intermediate object code
Speed	Fast execution	Slower execution
Errors	Displayed after the entire scan	Displayed after every lin